

Baseball

Note: For the purposes of this content, baseball programming will be a methodology shared across pitchers and position players. Although a case can be made for different programming strategies between pitchers and position players, clinical implementation has proven that the same OPT™ program can have tremendous impacts on injury prevention and performance enhancement for both pitchers and position players due to the overall demands of the sport.

Demands of the Sport (Gambetta, 1999)	
General information	• 4.3–4.4 seconds: Sprint time from home plate to first base • 3.6 seconds or less: Sprint time to steal from first base to second base • 9 minutes: Average time spent on defense for a half inning
Pitchers	• Starting pitcher: 6⅓ innings pitched per game • Nine-inning game: A pitcher will go nine innings 15% of all starts • Average number of pitches per inning: 15 • Average fastball velocity (first inning): 87 mph
Catchers	• Total throws: 221 • Throws from standing position: 89 • Throws from knees: 76 • Total sprints: 15 @ 30 yards/2 @ 5 yards • Total squat movements: 238 • Time on defense: 1 hour 40 minutes 50 seconds per game

Suggested Assessments for Baseball

Assessment Type	Assessment Method
Transitional assessments	• Overhead squat • Single-leg squat • Split squat • Star excursion balance test
Mobility assessments (if needed)	• Foot and ankle testing • Lumbo-pelvic-hip complex testing • Shoulder and spine testing
Dynamic movement assessments	• Drop jump assessment • Shark skill test • Davies test • Triple crossover hop test
Aerobic test: Choose one of the following	• 12-minute run test • 1.5-mile run test • Beep test
Anaerobic test	• 300-yard shuttle run
Muscular endurance assessments	• Push-up test • Pull-up test • Curl-up test
Muscular strength assessments	• 1–5 RM bench press • 1–5 RM squat • Nordic hamstring lower test
Power assessments	• Vertical jump test (double-leg) • Vertical jump test (single-leg) • Rotational medicine ball throw test (left and right)
Speed assessments (Norms may not exist for the following tests. Use these as objective measures of improvement over time.)	• 30-yard sprint test • Timed: Home to first base • Timed: Home to second base • Timed: Home to third base
Agility assessments	• LEFT (lower-extremity functional) test • 5–10–5 (pro agility test)

Off-Season Goals and Considerations for Baseball

Goals	Considerations
- Physical and mental reconditioning - Correcting impaired movement patterns based on results of transitional postural assessments - Improving: - Range of motion, joint stability, and muscle strength and endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total-body stabilization, stabilization, and power - Increasing: - Arm and shoulder strength and stability - Speed, agility, and quickness (position players)	- Planning and programming will be based on assessment, client goals, and the demand of the sport. - A systematic off-season plan will incorporate integrated training including: - Core training - Balance training - Plyometric training - Speed, agility, and quickness training - Integrated resistance training - Cardiorespiratory training combined with a proper nutrition plan to maximize training and recovery - An off-season training schedule will be developed by working backward from the commencement of spring training/season to allow for maximum injury prevention and performance gains without the athlete overtraining. - Performance programs will be modified appropriately, from a volume and exercise standpoint, to accommodate baseball-specific activities to avoid overtraining and maximize results. The athlete will continue to increase baseball skill activities.

Sample Off-Season OPT Baseball Plan (October–February)

Level	Phase	JAN	FEB	MAR	APR	MAY–SEP	OCT	NOV	DEC
		Off-Season (4)	Off-Season (5)	Pre-Season (4)	In-Season (5)	In-Season	Off-Season (1)	Off-Season (2)	Off-Season (3)
Corrective							X		
Stabilization	1		X					X	
Strength	2		X						X
	3								
	4								
Power	5	X	X						
	6								

Assessment & Program Strategies: Baseball



Sample Off-Season Weekly OPT Baseball Schedule

Month 1: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Corrective	OFF	Corrective	OFF	Corrective	OFF	OFF

Month 2: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Stabilization Endurance	OFF	Stabilization Endurance	OFF	Stabilization Endurance	OFF	OFF

Month 3: 4 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Corrective	OFF	Corrective	OFF	Corrective	OFF	OFF
Other	Conditioning	Conditioning			Conditioning		

Month 4: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Stabilization Endurance	OFF	Stabilization Endurance	OFF	Stabilization Endurance	OFF	OFF
Other		Conditioning		Conditioning			

Month 5: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Corrective	OFF	Corrective	OFF	Corrective	OFF	OFF
Other		Conditioning		Conditioning			

Conditioning = cardio training and/or sport-specific drills.

Sample Off-Season OPT Baseball Programs

Corrective Exercise Training

Core Exercises Training

INHIBIT		Sets	Reps	Time		
Myofascial Rolling						
Tensor fascia latae		1		30 sec		
IT band		1		30 sec		
Thoracic spine		1		30 sec		
Lats		1		30 sec		
LENGTHEN		Sets	Reps	Time		
Gastrocnemius stretch		1		30 sec		
Supine piriformis stretch		1		30 sec		
Posterior shoulder stretch		1		30 sec		
Lat stretch		1		30 sec		
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Side-lying hip abduction		2	10	4/2/1		0 sec
Floor bridge		2	10	4/2/1		0 sec
Standing shoulder external rotation		2	10	4/2/1		0 sec
Bent-over DB cobra		2	10	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Back	Single-leg cable row	2	10	4/2/1	60%	0 sec
Shoulders	Single-leg shoulder scaption	2	10	4/2/1	60%	90 sec

Assessment & Program Strategies: Baseball

Phase I: Stabilization Endurance Training

WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Lats			1		30 sec	
Static Stretching						
Gastrocnemius stretch			1		30 sec	
Supine piriformis stretch			1		30 sec	
Posterior shoulder stretch			1		30 sec	
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
Medicine ball woodchop		1	10	Moderate	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Quadruped arm opposite-leg raise		2	15	4/2/1	0 sec	
Floor bridge		2	15	4/2/1	0 sec	
Single-leg MB lift and chop		2	15	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
Box jump with stabilization		2	10	Hold landing	50 sec	
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo	Rest	
Speed ladder: Four exercises		2			60 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Step-up, balance to overhead press	2	15	4/2/1	65%	0 sec
Chest	SL suspension chest press	2	15	4/2/1	65%	0 sec
Back	Single-leg cable row	2	15	4/2/1	65%	0 sec
Shoulders	Single-leg scaption	2	15	4/2/1	65%	0 sec
Legs	Single-leg Romanian deadlift	2	15	4/2/1	65%	90 sec
COOL-DOWN						
Myofascial rolling and static stretching						

Assessment & Program Strategies: Baseball

Phase 2: Strength Endurance

WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
IT band			1		30 sec	
Thoracic spine			1		30 sec	
Lats			1		30 sec	
Active Stretching						
Standing adductor stretch			1	10		
Standing hip flexor stretch			1	10		
Floor lat stretch			1	10		
Dynamic Stretching		Sets	Reps	Tempo		
Multiplanar lunge with reach		1	10	Moderately fast	0 sec	
Squat to overhead reach		1	10	Moderately fast	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Cable rotation		2	10	1/1/1	0 sec	
Cable chop		2	10	1/1/1	0 sec	
Single-leg squat touchdown		2	10	1/1/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
Cone jumps: Side to side		3	10	Repeating	0 sec	
Power step-ups		3	10	Repeating	60 sec	
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance	Rest	
Speed ladder: Six exercises		2			60 sec	
T-drill		4		10 yd	60 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Optional					
Chest	Incline DB press	3	8	2/0/2	80%	0 sec
	Cable chest press		12	4/2/1	50%	
Back	Seated lat pull-down	3	8	2/0/2	80%	0 sec
	Single-leg cable row		12	4/2/1	50%	
Shoulders	Seated overhead DB press	3	8	2/0/2	80%	0 sec
	Bent-over DB combo II		12	4/2/1	50%	
Legs	Barbell squats	3	8	2/0/2	80%	0 sec
	Side lunge to balance		12	4/2/1	50%	60 sec
COOL-DOWN						
Myofascial rolling and static stretching						

Assessment & Program Strategies: Baseball

Phase 5: Power Training

WARM-UP			Sets	Reps	Time		
Myofascial Rolling							
Tensor fascia latae			1		30 sec		
Piriformis			1		30 sec		
Lats			1		30 sec		
Dynamic Warm-up (Some athletes may still require static stretching prior to dynamic warm-up.)							
Lateral tube walking			1	10	0 sec		
Lunge w/rotation			1	10	0 sec		
Push-up w/rotation			1	10	0 sec		
Iron cross			1	10	0 sec		
Russian twist			1	10	60 sec		
CORE, BALANCE, AND PLYOMETRICS			Sets	Reps	Tempo	Rest	
No CORE, BALANCE, AND PLYOMETRICS (incorporated into dynamic warm-up and resistance training)							
SPEED/AGILITY/QUICKNESS			Sets	Reps	Distance	Rest	
Speed ladder: Six exercises			2			60 sec	
T-drill			4		10 yd	60 sec	
RESISTANCE			Sets	Reps	Tempo	Intensity	Rest
Total body	Optional						
Chest	Incline DB press	3	5	Controlled	85%	0 sec	
	Rotational chest pass		10	Explosive	2% of BW	2 min	
Back	Seated cable row	3	5	Controlled	85%	0 sec	
	MB soccer throw		10	Explosive	2% of BW	2 min	
Shoulders	Shoulder press machine	3	5	Controlled	85%	0 sec	
	Overhead MB throw		10	Explosive	2% of BW	2 min	
Legs	Barbell squat	3	5	Controlled	85%	0 sec	
	Ice skater		10	Explosive	2% of BW	3 min	
COOL-DOWN							
Myofascial rolling and static stretching							

Preseason Goals and Considerations for Baseball

Goals	Considerations
- Correcting impaired movement patterns based on results of preseason assessments - Maintaining: - Physical health (decrease chance of injury for participation in practice and games, and optimal performance within such activities) - Range of motion, joint stability, and muscle strength and endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total-body stabilization, strength, and power - Speed, agility, and quickness (position players)	- A systematic preseason plan that transitions the athlete from performance enhancement to injury prevention and maintenance of off-season gains will be developed. This transition should be implemented with the consideration that the athlete's sport participation level is increasing dramatically from what they have become used to in the off-season. The proper recovery and conditioning should be applied during this time of increased activity. - Consider the exercise volume of injury prevention and performance training based on the athlete's increased activity level.

Sample Preseason OPT Baseball Plan (March)

Level	Phase	JAN	FEB	MAR	APR	MAY-SEP	OCT	NOV	DEC
		Off-Season	Off-Season	Pre-Season (1)	In-Season	In-Season	Off-Season	Off-Season	Off-Season
Corrective				X					
Stabilization	1			X					
Strength	2			X					
	3								
	4								
Power	5								
	6								

Sample Preseason Weekly OPT Baseball Schedule

Month 1

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance	Corrective	OFF	Stabilization Endurance	Corrective	OFF	OFF
Other	Game	Game	Game		Game	Game	

Sample Preseason OPT Baseball Programs

Corrective Exercise Training

INHIBIT				Sets	Reps	Time	
Myofascial Rolling							
Tensor fascia latae				1		30 sec	
Piriformis				1		30 sec	
Lats				1		30 sec	
LENGTHEN				Sets	Reps	Time	
Gastrocnemius stretch				1		30 sec	
Supine biceps femoris stretch				1		30 sec	
Standing pec stretch				1		30 sec	
Floor lat stretch				1		30 sec	
ACTIVATE			Sets	Reps	Tempo	Intensity	Rest
Single-leg calf raise			2	10	4/2/1		0 sec
Standing shoulder external rotation			2	10	4/2/1		0 sec
Bent-over DB combo I			2	10	4/2/1		0 sec
INTEGRATE			Sets	Reps	Tempo	Intensity	Rest
Back	Single-leg, single-arm cable row		2	12	4/2/1	60%	0 sec
Shoulders	Single-leg, single-arm scaption		2	12	4/2/1	60%	90 sec

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WARM-UP				Sets	Reps	Time
Myofascial Rolling						
Tensor fascia latae				1		30 sec
Piriformis				1		30 sec
Lats				1		30 sec
Static Stretching						
Gastrocnemius stretch				1		30 sec
Supine biceps femoris stretch				1		30 sec
Posterior shoulder stretch				1		30 sec
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate		0 sec
Medicine ball woodchop		1	10	Moderate		0 sec
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Side iso-abs with hip abduction		2	15	4/2/1	0 sec	
Floor cobra		2	15	4/2/1	0 sec	
Single-leg balance reach		2	15	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS because of increased activity						
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance	Rest	
No SAQ because of on-field drills and conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Optional	2	15	4/2/1	65%	0 sec
Chest	Push-up: Feet in suspension	2	15	4/2/1	65%	0 sec
Back	SL suspension row	2	15	4/2/1	65%	0 sec
Shoulders	Single-leg scaption	2	15	4/2/1	65%	0 sec
Legs	Step-up to balance: Transverse plane	2	15	4/2/1	65%	90 sec
COOL-DOWN						
Myofascial rolling and static stretching						

Assessment & Program Strategies: Baseball

Phase 2: Strength Endurance Training

WARM-UP		Sets	Reps	Time		
Myofascial Rolling						
Piriformis		1		30 sec		
Adductors		1		30 sec		
Thoracic spine		1		30 sec		
Active Stretching						
Gastrocnemius stretch		1	10			
Standing adductor stretch		1	10			
Standing pec stretch		1	10			
Dynamic Stretching		Sets	Reps	Tempo	Rest	
Multiplanar lunge with reach		1	10	Moderately fast	0 sec	
Squat to overhead reach		1	10	Moderately fast	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Cable crunch		2	10	1/1/1	0 sec	
Back extension		2	10	1/1/1	0 sec	
Step-up to balance: Transverse plane		2	10	1/1/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS because of increased activity						
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance	Rest	
No SAQ because of on-field drills and conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Optional					
Chest	Machine chest press	3	12	2/0/2	70%	0 sec
	Push-up: Feet in suspension		8	4/2/1		
Back	Seated cable row	3	12	2/0/2	70%	0 sec
	Back extension		8	4/2/1		
Shoulders	Standing overhead press	3	12	2/0/2	70%	0 sec
	SL DB scaption		8	4/2/1		
Legs	Forward DB lunge	3	12	2/0/2	70%	0 sec
	Single-leg squat		8	4/2/1		60 sec
COOL-DOWN						
Myofascial rolling and static stretching						

In-Season Goals and Considerations for Baseball

Goals	Considerations
- Maintaining: - Range of motion, joint stability, and muscle strength and endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total-body stabilization, strength, and power - Speed, agility, and quickness (position players) - Arm and shoulder strength - Establishing with the athlete a communication process for monitoring in-season maintenance, and a plan that works based on what the athlete needs, wants, and will have the capability to do while playing baseball	- Assessments and testing, to ensure program considerations are being made to prevent overuse injuries and maintain performance levels, will be performed. - Impaired movement patterns with exercise strategies being implemented will be attended to. - A systematic in-season maintenance plan will be developed that allows the athlete to minimize decreases in performance and chance of injury over a long season with repetitive sport movement patterns (hitting and throwing). - The exercise volume of injury prevention, and performance training based on the athlete's increased travel and activity level, will be considered.

In-Season OPT Baseball Plan (April–September)

Level	Phase	JAN – FEB	MAR	APR	MAY	JUNE	JUL	AUG	SEPT	OCT–DEC
		Off-Season	Pre-Season	In-Season (1)	In-Season (2)	In-Season (3)	In-Season (4)	In-Season (5)	In-Season (6)	Off-Season
Corrective				X	X	X	X	X	X	
Stabilization	1			X	X	X	X	X	X	
Strength	2			X	X	X	X	X	X	
	3									
	4									
Power	5									
	6									

Sample In-Season Weekly OPT Baseball Schedule

Any Month

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance	OFF	Corrective	OFF	Stabilization Endurance	Corrective	OFF
Other	Game	Game	Game		Game	Game	Game

Sample In-Season OPT Baseball Program

Corrective Exercise Training

INHIBIT				Sets	Reps	Time
Myofascial Rolling						
Tensor fascia latae				1		30 sec
Piriformis				1		30 sec
Lats				1		30 sec
LENGTHEN				Sets	Reps	Time
Gastrocnemius stretch				1		30 sec
Supine biceps femoris stretch				1		30 sec
Standing pec stretch				1		30 sec
Floor lat stretch				1		30 sec
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Single-leg hip extension		1	15	4/2/1		0 sec
Floor bridge		1	15	4/2/1		0 sec
SL bent-over DB cobra		1	15	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Total body	Cable squat to row	3	10	4/2/1	60%	0 sec
Total body	Side lunge, curl to overhead press: Single-arm	3	10	4/2/1	60%	90 sec

Assessment & Program Strategies: Baseball

Phase I: Stabilization Endurance Training

WARM-UP				Sets	Reps	Time	
Myofascial Rolling							
Tensor fascia latae				1		30 sec	
Piriformis				1		30 sec	
Lats				1		30 sec	
Static Stretching							
Gastrocnemius stretch				1		30 sec	
Standing adductor stretch				1		30 sec	
Standing pec stretch				1		30 sec	
Dynamic Stretching			Sets	Reps	Tempo		
Lunge with rotation			1	10	Moderate	0 sec	
Medicine ball woodchop			1	10	Moderate	0 sec	
CORE AND BALANCE			Sets	Reps	Tempo	Rest	
Side iso-abs			2	15	4/2/1	0 sec	
Floor bridge: Single-leg			2	15	4/2/1	0 sec	
Single-leg balance reach			2	15	4/2/1	60 sec	
PLYOMETRICS			Sets	Reps	Tempo	Rest	
Multiplanar hop with stabilization box jump-down			2	10	Hold landing	60 sec	
RESISTANCE			Sets	Reps	Tempo	Intensity	Rest
Total body	Cable squat to row		2	15	4/2/1	65%	0 sec
Chest	Suspension push-up: Feet in straps		2	15	4/2/1	65%	0 sec
Back	SL bent-over DB Cobra		2	15	4/2/1	65%	0 sec
Shoulders	Single-leg cable lift		2	15	4/2/1	65%	0 sec
Legs	Step-up to balance: Frontal plane		2	15	4/2/1	65%	90 sec
COOL-DOWN							
Myofascial rolling and static stretching							

Assessment & Program Strategies: Baseball

Phase 2: Strength Endurance Training

WARM-UP		Sets	Reps	Time
Myofascial Rolling				
Tensor fascia latae		1		30 sec
IT band		1		30 sec
Lats		1		30 sec
Active Stretching				
Gastrocnemius stretch		1	10	
Standing adductor stretch		1	10	
Standing pec stretch		1	10	
Dynamic Stretching		Sets	Reps	Tempo
Multiplanar lunge with reach		1	10	Moderate
Squat to overhead reach		1	10	Moderate
CORE AND BALANCE				
Reverse crunch		3	10	1/1/1
Back extension		3	10	1/1/1
Turning step-up to balance		3	10	1/1/1
PLYOMETRICS		Sets	Reps	Tempo
Front power step-up		3	10	Repeating
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance
Speed ladder: Six exercises		2		60 sec
RESISTANCE		Sets	Reps	Tempo
Total body	Optional			
Chest	Incline DB press	2	10	2/0/2
	Cable chest press		10	4/2/1
Back	Seated cable row	2	10	2/0/2
	SL lat pull-down		10	4/2/1
Shoulders	Seated DB scaption	2	10	2/0/2
	SL SA DB press		10	4/2/1
Legs	Deadlift	2	10	2/0/2
	Reverse lunge to balance		10	4/2/1
COOL-DOWN				
Myofascial rolling and static stretching				